Percent and Ratio Reasoning

CIVIL-SERVICE · numerical-ability · BoardPrep lesson v1

Official examination scope

CSC Examination Announcement No. 03 s. 2026 lists basic operations, number sequence, and word problems under Numerical Ability. This lesson practices percent and ratio applications within basic operations and word problems. It does not estimate item frequency or section weight; check current CSC notices before an exam.

Learning objectives

- Identify the original whole, part, and rate before calculating a percent.
- Compute a percent of an amount and a percent change using the correct base.
- Solve a proportion or rate question and check that units and totals make sense.

1. Percent of a base

Percent means per hundred. Convert p percent to p/100, then multiply by the stated base. The base is the whole that the percent describes. For example, 18% of 250 is 0.18 x 250. Read the question for the whole before choosing an operation.

2. Percent change

Find the change first: new amount minus original amount. Divide that change by the original amount, then multiply by 100%. A decrease can be reported as a positive decrease when the question asks for its magnitude. The original amount remains the denominator because it is the comparison base.

3. Ratios and rates

An equivalent ratio scales every part by the same factor. A unit rate divides the measured quantity by the reference units, such as forms per hour. Convert units to a common basis before comparing rates. Keep the unit beside each intermediate value so a reversed ratio or mismatched time period is easier to catch.

Essential relationships

- part = rate as a decimal x base
- percent change = (new - original) / original x 100%
- equivalent ratio: multiply or divide every term by the same nonzero factor
- unit rate = quantity / reference units

Find a percent of an amount

Question: What is 18% of 250?
1. Write 18% as 0.18.
2. Multiply 0.18 x 250 = 45.
Answer: 18% of 250 is 45.
Check: Ten percent of 250 is 25 and eight percent is 20; 25 + 20 = 45.

Find a percent increase

Question: An amount rises from 500 to 575. What is the percent increase?
1. Change = 575 - 500 = 75.
2. Divide by the original amount: 75 / 500 = 0.15.
3. Convert to a percent: 0.15 x 100% = 15%.
Answer: The increase is 15% of the original amount.
Check: 500 x 1.15 = 575. Using 575 as the denominator would answer a different question.

Undo a discount

Question: A price after a 15% discount is 680. What was the price before the discount?
1. The sale price is 100% - 15% = 85% of the original.
2. Write 85% as 0.85 and solve original x 0.85 = 680.
3. Original = 680 / 0.85 = 800.
Answer: The original price was 800.
Check: A 15% discount on 800 is 120; 800 - 120 = 680. Adding 15% of 680 would use the wrong base.

Common misconceptions

- Using 18 instead of 0.18 when multiplying by a percent.
- Dividing percent change by the new amount instead of the original base.
- Adding a discount percent to the discounted price to reverse the discount.
- Scaling only one term of a ratio or comparing rates with different time units.

Practice patterns

- Compute a stated percent of a supplied base.
- Find a percent increase or decrease from an original and a new amount.
- Recover an original amount from a discounted or retained percentage.
- Scale a ratio, find a missing part, or compare unit rates after matching units.

Official scope references

Official scope: CSC Examination Announcement No. 03 s. 2026: 09 August 2026 CSE-PPT Scope — Philippine Civil Service Commission — https://csc.gov.ph/phocadownload/userupload/erpo/announcements/2026/ExamAnn_2026_03_Conduct%20of%2009%20Aug%202026%20CSE-PPT_orig%20signed.pdf

Instructional references

Instructional reference: Prealgebra 2e, Section 6.2: Solve General Applications of Percent — OpenStax, Rice University — https://openstax.org/books/prealgebra-2e/pages/6-2-solve-general-applications-of-percent
Instructional reference: Prealgebra 2e, Section 6.5: Solve Proportions and Their Applications — OpenStax — https://openstax.org/books/prealgebra-2e/pages/6-5-solve-proportions-and-their-applications

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